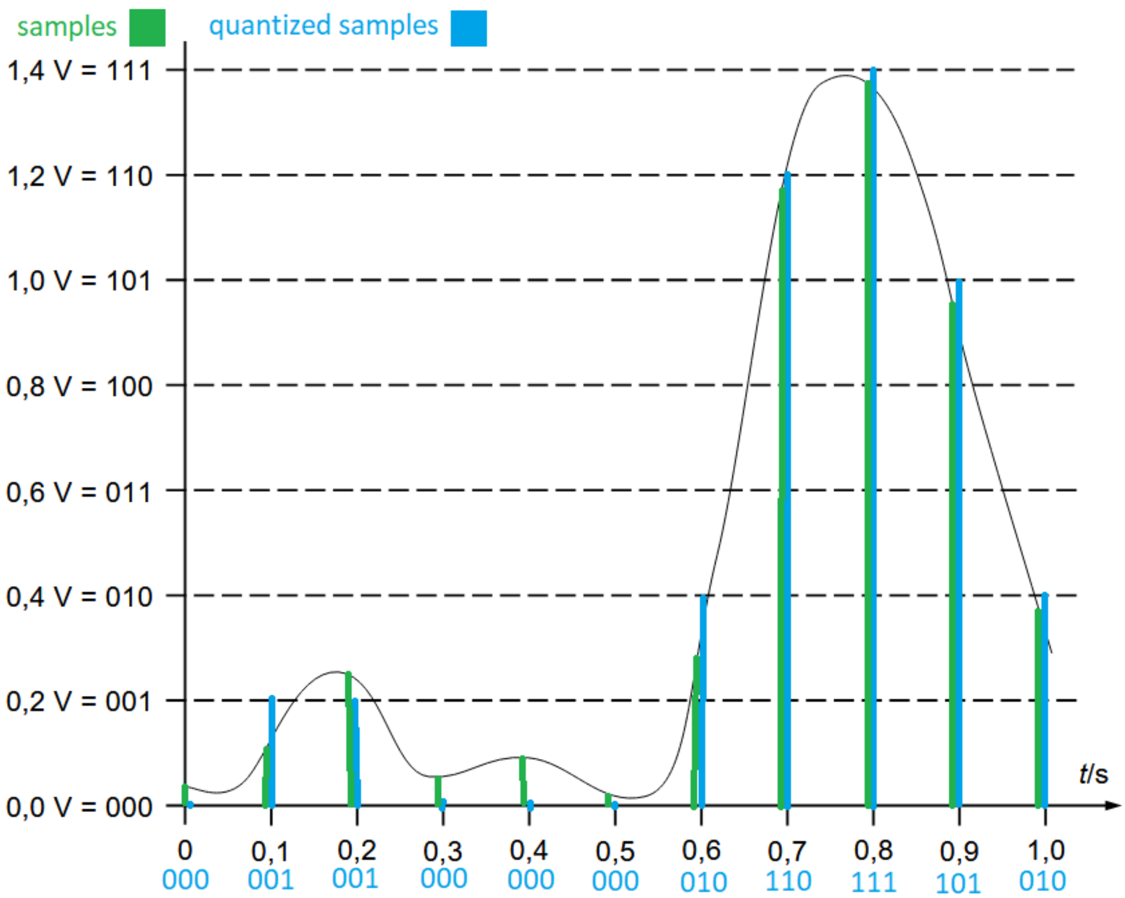


```
2.1
    2^12 - 1 = 4095
    1.
        (2734 / 4095) * 3,3 V = ~2,203 V
    2.
        (3999 / 4095) * 3,3 V = ~3,223 V
    3.
        (137 / 4095) * 3,3 V = ~0,110 V
    4.
        (512 / 4095) * 3,3 V = ~0,413 V
2.2
    2^14 - 1 = 16383
    1.
        (3,3 V / 5 V) * 16383 = 10818,78
    2.
        (0,65 V / 5 V) * 16383 = 2129,79
    3.
        (5,0 V / 5 V) * 16383 = 16383
    4.
        (2,1 V / 5 V) * 16383 = 6880,86
3.1
    see attached python file*
3.2
    the trimmer outputs an analog signal that is read in python, and is then divided by the maximum value to calculate
    how much of a single second the program will wait until reversing the state of the led light.
    when the trimmer is turned all the way counter clockwise it is in its maximal output and the light has a delay of 1 second.
    and when turned fully clockwise it flips the led as fast as the program can run.
```